Lab 28 — NumPy Indexing and Filtering

Aim

To demonstrate indexing, slicing, and Boolean filtering on multi-dimensional NumPy arrays.

Algorithm / Procedure

1. Import NumPy.
2. Create an array using arange() and reshape it into a 2D array.
3. Access a specific element using indexing.
4. Extract a column using slicing.
5. Filter elements using a Boolean condition.
6. Replace even elements with zero using Boolean indexing.
7. Display the results.

Program

```
import numpy as np

a=np.arange(12).reshape(3,4)

print("Array:\n",a)

print("a[1,2]:",a[1,2])

print("Column 1:",a[:,1])

print("Elements > 5:",a[a>5])

a[a%2==0]=0

print("After zeroing evens:\n",a)
```

Output

```
Array:
 [[ 0  1  2  3]
 [ 4  5  6  7]
 [ 8  9 10 11]]
a[1,2]: 6
Column 1: [1 5 9]
Elements > 5: [ 6  7  8  9 10 11]
After zeroing evens:
 [[ 0  1  0  3]
 [ 0  5  0  7]
 [ 0  9  0 11]]
```

Result

NumPy indexing, slicing, Boolean filtering, and conditional assignment operations were successfully demonstrated on a multi-dimensional array.